Pollution of the Oceans

- Oceans draw the boundaries of many nations
- Oceans have a big say in the local and global climate

- UN declared 1998 as the Year of Ocean with an objective to focus and reinforce the attention of the public, governments, etc on the importance of oceans and marine environment as resources for sustainable development.

Importance of Ocean

- About 70% of earth's surface is covered by ocean / marine environment
- Life took birth in ocean (present wisdom)
- Vast resources of the oceans: Food (marine products), Energy, Drugs………
- Ocean is also used for navigation, waste disposal, testing of nuclear weapons, etc.

- Man knows only very little about ocean
- Only about 1% of the ocean and the ocean bed has been explored so far.

Ocean and Global Climate

- There is a strong link between the ocean and the climate
- Ocean circulations are heavy and slow compared to that of the atmosphere
- But heat capacity of water (specific heat 4.18 kJ/kg ^{o}C at 27^{o}C) is much larger than that of the air (1 kJ/kg ^{o}C at 27^{o}C)
- Hence movement of heat across the oceans is very significant in the climate system

- On the average, complete mixing of water in oceans take place once in 2000 years
- Oceans transport massive amount of heat around the planet
- Heat transport takes place due to the temperature and salinity (hence density) difference in sea water.
- If the above thermohaline circulation breaks down, there could be serious consequences
- ElNino is an occasional incursion of warm water along the coast of Peru. It is considered to be causing climatic changes

- 1997 El Nino was reported to cause severe drought in India, Indonesia, etc. and heavy showers and floods in Uruguay, Brazil, Chile, Peru, etc.
- But in India, there was no any clear link between El Nino and Indian monsoon.
- But there was a harsh hot spell in Indonesia causing massive forest fires

Productivity of Oceans

- Ocean is biologically very active
- Arabian sea is one of the most biologically productive ocean regions
- That is mainly due to the up welling of nutrients during summer
    - *Biological activity of oceans is measured by lowering a white disc called Secchi. The depth at which it becomes invisible is noted. More the phytoplanktons (productivity), the sooner the disc disappears from the view*

- Over geological time, photosynthetic carbon fixation in the oceans has exceeded the respiratory oxidation of organic carbon
- This imbalance between the two processes resulted in the simultaneous accumulation of Oxygen in the atmosphere, draw down of CO_2 from the atmosphere and final burial of organic carbon in the marine sediments
- Nitrogen cycle (not P cycle as it was thought before) controls the variation of CO_2 in the atmosphere

- Nitrogen fixation governs the primary productivity on geological scale
- Small variations in the ratio of nitrogen fixation to denitrification (due to changes in the oxidation state of oceans) can significantly change the atmospheric CO_2 concentration on a geological scale.
- Once upon a time, N was a rare precious commodity
- Only specialized bacteria and lightning could convert atmospheric N into the biologically usable forms.

- But today N is available in plenty (due to the human intervention) in the biological world
- Now humans control more N than all the natural processes combined together
- About 140 million tones of N per year is released from anthropogenic sources into the atmosphere.
- Ocean is the largest sink of CO_2
- It absorbs about 7.3 billion tones of CO_2 per year

Recourses of Oceans

- Food (fish and other marine products)
- Medicines and drugs *- numerous*
- CH_4 hydrates for methane gas
- Liquid petroleum and natural gas
- Tidal energy
- Wave energy (10 to 100 kw per meter length of the wave).
- temperature difference between the surface and the depths

The Problems ….MARINE POLLUTION

- Oil Spills
- Hazardous waste trafficking, accidents and dumping
- Waste (MSW, Sewage, nuclear waste, CO_2, etc) disposal
- Tourism
- Fishing
- Warfare (underwater mines, submarines, etc.) & Nuclear experiments
- Algal blooms

Oil Slick

- Very frequent
- It directly affects the marine lives and sea birds
- More dangers if washed ashore
  - Polluting kilometers of beach
  - Damaging intake structures power plants
  - Damaging the coastal ecosystem

OIL SLICKS INCIDENTS

- On Jan. 19, 1991 in Persian Gulf, Kuwait **Amount spilled:** 380-520 million gallons
- This worst oil spill in history wasn't an accident — it was deliberate. During the Gulf War, Iraqi forces attempted to prevent American soldiers from landing by opening valves at an offshore oil terminal and dumping oil from tankers. The oil resulted in a 4-inch thick oil slick that spread across 4,000 square miles in the Persian Gulf.

- April 22, 2010 in the Gulf of Mexico
- **Amount spilled:** An estimated 206 million gallons
- This oil spill is officially the largest accidental spill in world history.

- on June 3, 1979 in the Bay of Campeche off Ciudad del Carmen, Mexico
  Amount spilled: 140 million gallons

- On July 19, 1979 Off the coast of Trinidad and Tobago: 90 million gallons
- On Feb. 10, 1983 in the Persian Gulf, Iran : 80 million gallons
- On Aug. 6, 1983 in the Saldanha Bay, South Africa: 79 million gallons

- On March 16, 1978 at Portsall, France: 69 million gallons
- On May 28, 1991 About 700 nautical miles off the coast of Angola : 51-81 million gallons
- On April 11, 1991 at Genoa, Italy: 45 million gallons
- **On** Nov. 10, 1988 Off the coast of Nova Scotia, Canada: 40.7 million gallons

- On Dec. 19, 1972 in the Gulf of Oman: 35.3 million gallons
- On March 18, 1967 in Sicilly Isles, UK: 25-36 million gallons

A Few Incidents of 1997

- 1997: Nakhodka (Russian tanker) carrying about 19 million tones of fuel oil spilt open in a storm on 2nd January. Large oil slick drifted towards Wakasa bay, Japan. Wakasa is the site of intakes of many nuclear power stations of Japan including Monju. If it had hit Wakasa, 5% of Japan's electric power would have been curtailed

- 1997: 2nd July: Diamond Grace ran aground in Tokyo bay spilling 3.90 million gallons of crude oil off the coast of Yokohama.
- 1997: October 15th: Oil tanker Evoikos (Cyprus registered) and Orpin Global (Thai) collided in Singapore strait spilling about 7 million gallons of fuel oil. The main island of Singapore had a narrow escape.
- 1997: June 19th: MV Arcadia Pride (Indian) capsized near Bombay coast spilling oil. This would have caused problems in Bombay, but drifted away.

Prestige and Erika Oil Spills

- 2002/2003 two infamous oil slicks in the Atlantic ocean
- Oil slicked to the seashore of many countries
- For instance: 450 km of French coast was contaminated

Fighting Oil Slick

- Spraying chemical dispersants to break up and dissolve the oil slick. This enhances the degradation rate of oil.
  - Chemical dispersants are most effective on fresh oil (Oil : Dispersants = 3:1)
  - For old oil slick, viscous / heavy oil, more dispersants are required (Oil : Dispersants = 1:1)

- Collect oil mixed water and recover oil from that: protective fencing, absorbent mats are all used for this.

- Burning: if the oil slick is fresh, thick and continuous.
    - Wicking agents needed
    - Not advisable
- Sinking: distribution of a dense material over the oil slick to sink it down to the bottom of the ocean. Sand treated to increase its affinity for oil can be used.
    - 1 ton oil needs about 1 tone of sand
    - Not recommended for heavily fished area
    - May foul the fishing nets and harm flora and fauna

Algal Bloom – Red Tides

- Caused by the bloom of Dinoflagellates and some species of algae.
- Warmer seas and increasing levels of pollution are responsible
- Red tides last only for 2-3 weeks.
- Marine lives gulp the toxins produced by these planktons
- Red tides can block oxygen supply and sun light to the marine lives.

Control of Red Tides

- Practically no direct control of marine bloom is possible.
- Keep coastal ecosystem clean
- Biological control using predators like zooplanktons, viruses, etc. – untested
- Combining biological control with chemical agents – un explored
- Use of chemical agents: copper sulphate (not very effective)

- Use of coagulants: clay. The bloom is carried down to the bed.
  - Use of coagulant aid like polyhydroxi aluminium chloride shall increase the efficiency of clay

- Ocean is a heritage for the future
- The exact value of ocean to the mankind is still not clear !!
- Without healthy oceans, earth's life supporting system cannot work efficiently.
- Marine pollution is to be taken seriously.